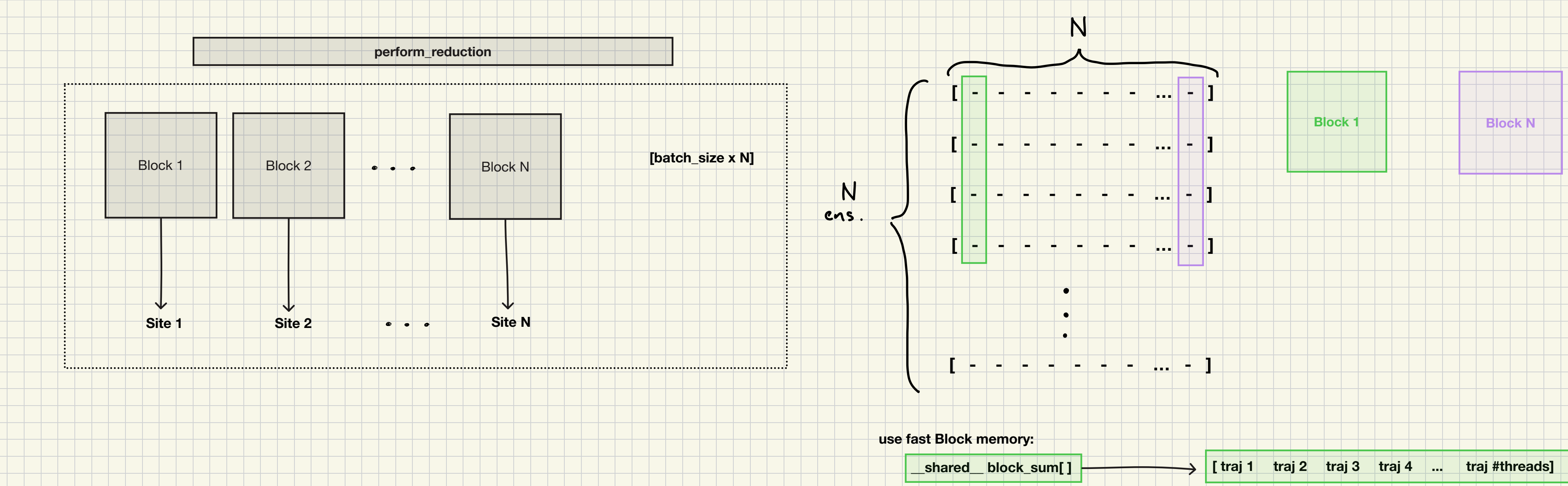
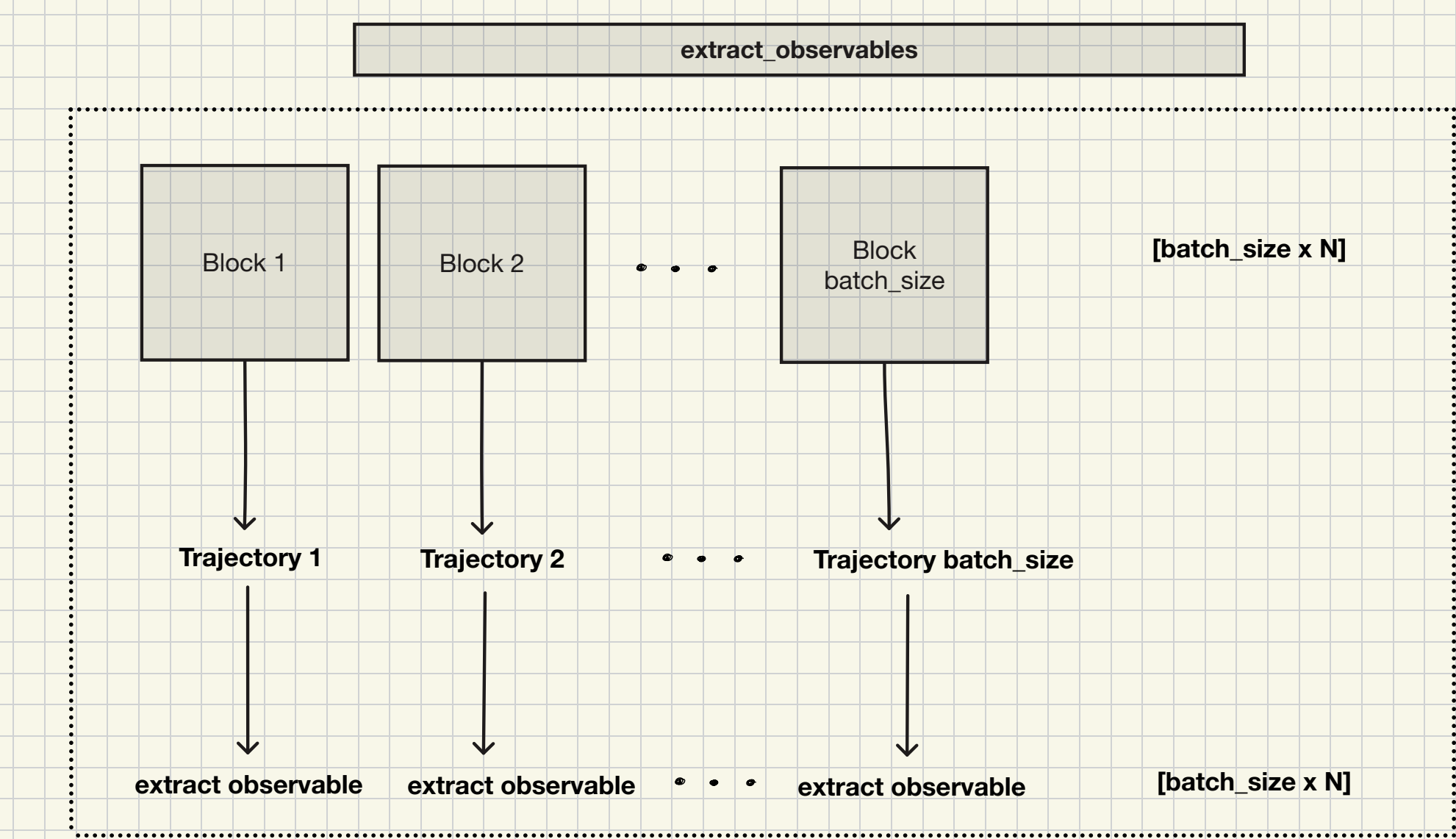


Batch 2



to avoid `atomicAdd()`:
use tree based add:

#threads / 2 = 128

$$\left. \begin{array}{l} \text{batch_sum}[x_1] * \text{batch_sum}[x_1] + \text{batch_sum}[x_1 + x_2] \\ \text{batch_sum}[x_1] * \text{batch_sum}[x_2] + \text{batch_sum}[x_2 + x_2] \\ \text{batch_sum}[x_3] * \text{batch_sum}[x_3] + \text{batch_sum}[x_3 + x_2] \\ \vdots \\ \text{batch_sum}[x_n] * \text{batch_sum}[x_n] + \text{batch_sum}[x_n + x_2] \end{array} \right\} \rightarrow \dots$$

2 / 2 = 1

$$\left. \begin{array}{l} \text{batch_sum}[x_1] * \text{batch_sum}[x_1] + \text{batch_sum}[x_1 + 1] \\ \text{batch_sum}[x_1] * \text{batch_sum}[x_2] + \text{batch_sum}[x_2 + 1] \end{array} \right\} \rightarrow \dots$$

this is $\log_2(\#threads)$ instead of expensive $\#threads$